TEAM VBALL · SDC HACKATHON 2026 · FINAL ROUND · DOMAIN 2

SMErgers

Agreed on WhatsApp. Billed from memory.
Not any more.

01
Catch it

02
Confirm it

03
Sync it

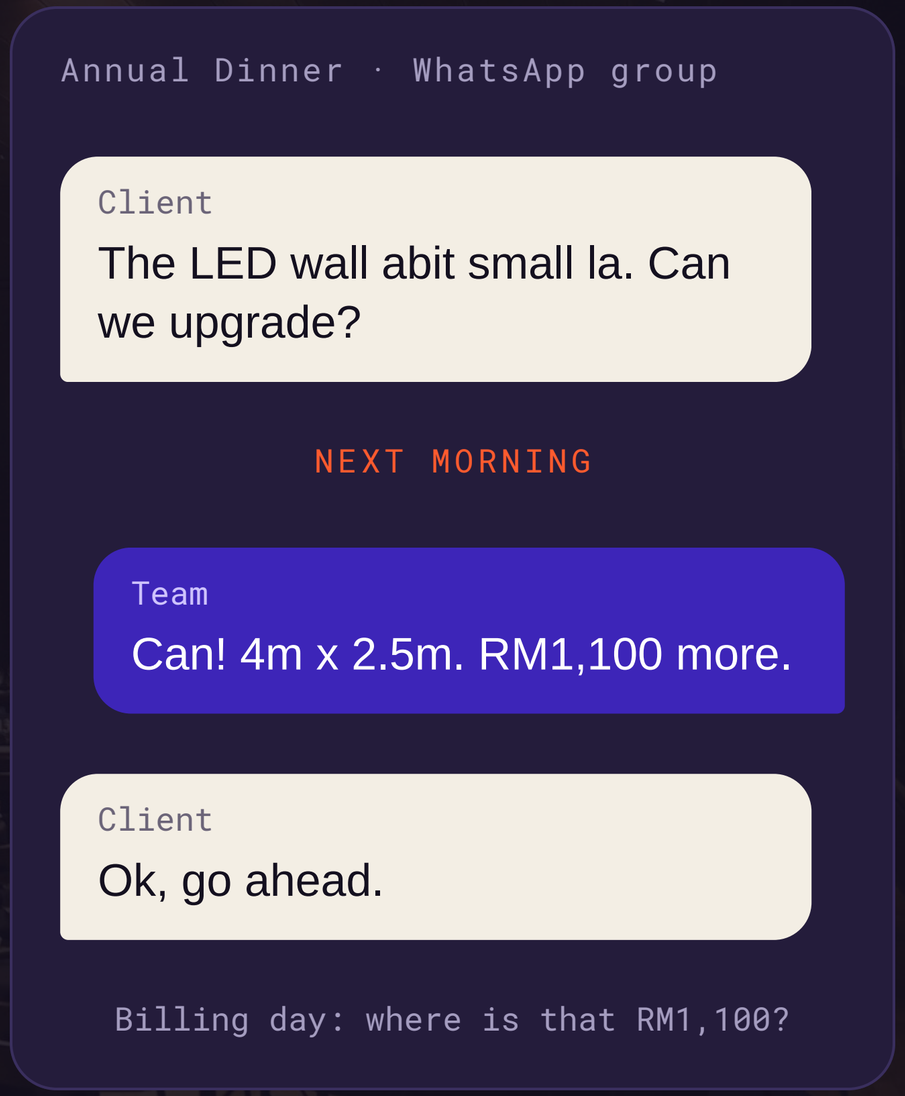
Orbyte Smart SME Challenge
SDG 8.2 · 8.3

PROBLEM

Agreed on WhatsApp. *Lost at billing.*

1. The client asks in the chat.
2. The team says yes.
3. The signed quote is never updated.

So the extras are billed from memory, *or not at all.*

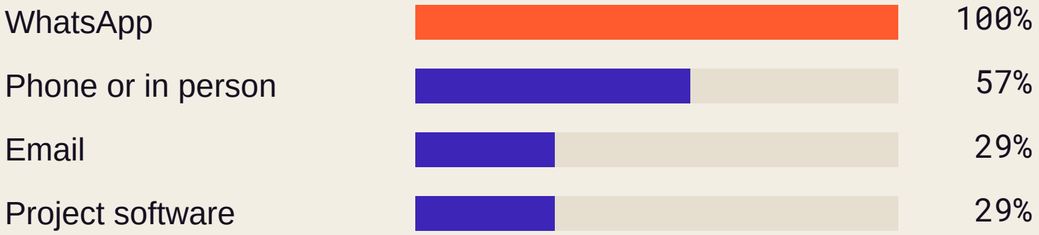


Illustrative example: a synthetic dinner job, written from a team member's experience at an event company.

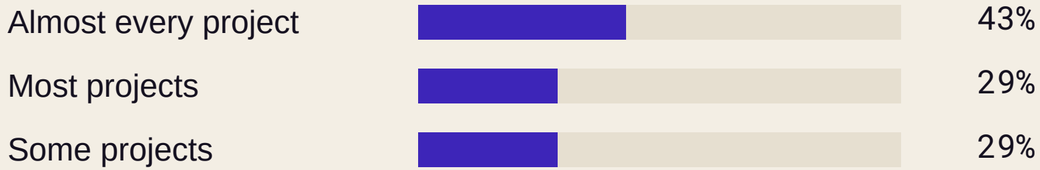
SURVEY · 7 EVENT PROFESSIONALS · 5 COMPANIES

We surveyed 7 event professionals *at 5 companies.*

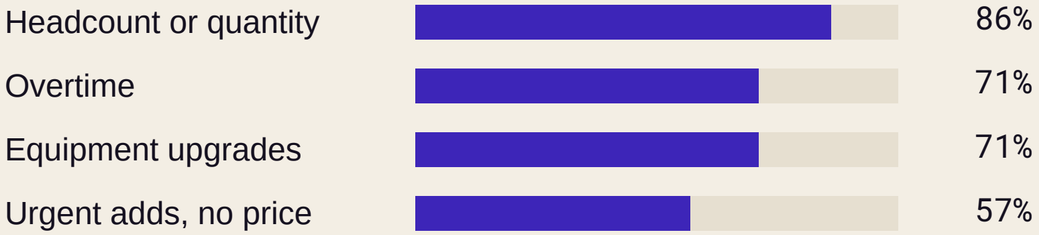
WHERE CHANGES HAPPEN



CLIENTS ADD EXTRAS AFTER SIGNING



WHAT THEY ADD



QUOTE VALUE LOST ON A TYPICAL JOB



Nobody said under 6%.

Survey of 7 people at 5 event companies, 31 Oct 2026. Our network, self-reported estimates, small sample. Table: docs/survey.md.

PROBLEM

Where the money disappears

01

Quote signed

RM34,530

Sample dinner quote, 180 guests

02

Changes in WhatsApp

100%

of the 7 we asked run their changes in WhatsApp

03

Bill from memory

57%

scroll back through weeks of chat to find them

04

“I never agreed.”

“...their boss doesn’t admit and doesn’t want to pay.”

Owner, 100+ jobs a year

05

Owner gives in

71%

lose 11% or more of a typical job’s value

Nobody is careless. The agreement just lives in *a chat, not a system.*

Survey of 7 people at 5 event companies, 31 Oct 2026, self-reported. The quote in step 1 is a sample.

Competitive Analysis

Nobody turns the chat into the bill

CAN IT…	WhatsApp + memory	Accounting apps Bukku, AutoCount	Event software Rentman, Event Temple	Contractor tools CoConstruct, Baukraft	SMErgers
Read agreements from WhatsApp	×	×	×	Partly	✓
Match each change to the quote	×	×	Partly	✓	✓
Client confirms each change	×	×	Partly	✓	✓
Post the new version to the group	×	×	×	×	✓
Link each bill line to its chat proof	×	×	×	×	✓

Public pages and search results, 21 Oct 2026. Partly = a limited form. × = not found, not proof it does not exist.

SOLUTION

Catch it. *Confirm it.* Sync it.

CATCH IT



Messages arrive

From the job's WhatsApp group



Chat analysed

Reader finds item, quantity, price, source

CONFIRM IT



PM checks

Adds any missing price



Director signs off

RM500 or more, and discounts



Client confirms

From a link, no login

SYNC IT



Quote v2 to the group

Plus bill, CSV, log

In our test, all 15 agreed changes reached the bill, with a person checking each one.

Synthetic test: 4 unseen chats, 15 agreed changes. Reproduce: python src/benefit.py

LIVE DEMO

Watch it *catch a change.*

A real WhatsApp group, from the owner’s chair.

Test numbers and sample business data. Backup: the recorded demo.

TECHNICAL APPROACH

The AI reads. *Code and people decide.*

TRIGGER	DATA	CHAT ANALYSIS	APPROVAL	OUTCOME
WhatsApp group A message arrives	**Quote and rate card** Quote lines, prices, who’s who	**Reads the chat** Rules at once. Llama AI reads the chat against the quote	**PM, director, client** Roles enforced on the server	**Quote v2 and bill** Posted back to the group

If the AI is down, the rules keep working.

Labels are the five steps of the Domain 2 brief. AI reader: open Llama 3.1 8B on a private GPU endpoint. Accuracy results: appendix.

ORBYTE INTEGRATION

Plugs into Orbyte. *Adds what it lacks.*

SMErgers needs	Orbyte provides
The job and its client	Clients and projects
Who is PM, director, client	Directory, shared roles
Director and client sign-off	Approvals, eSign
A record nobody can edit	Audit trail
Sending the bill out	Accounting partners
Job data for other modules	Open APIs

WHAT WE ADD

Chat turned into approved changes

1 **TODAY**
Job API and the official WhatsApp webhook format

2 **PILOT**
Use Orbyte’s roles, approvals and eSign instead of ours

3 **ADD-ON**
One switch in Orbyte for SMEs that quote jobs

Orbyte modules as in the organizers’ introduction. Designed, not yet connected to a live Orbyte: docs/orbyte-integration.md.

WHO WE SELL TO

Start with events. *Then any firm that quotes a job.*

PRIMARY · FIRST TARGET

Small event companies

Quote signed, then changes in WhatsApp. This is where our survey and our demo are.

SECONDARY · NEXT, WITH A NEW QUOTE FILE

- Renovation and fit-out — next
- Air-con and repair — next
- Catering — next
- Printing — next
- Web and creative agencies — next
- Wholesale and supply orders — next

Secondary markets are planned, not yet piloted. Each is a new quote file for the same engine.

BUSINESS MODEL

We sell software by the month. *The owner pays.*

TRY

Free

1 job at a time

RUN

RM100 a month

10 jobs at a time

SCALE

RM250 a month

Unlimited jobs

---

5 of 7 event pros we asked lose *11%+ of a typical job.* 6 of 7 picked a paid plan as fair.

Plans and limits are a launch proposal; the prices came from our survey of 7. How much an owner recovers is not measured yet: the pilot measures it.

MARKET SIZE

How big? *Counted bottom-up.*



TAM · EVERY FIRM

About 128,000 firms

Small firms that run projects in Malaysia, at RM1,200 a year each

SAM · THE ONES WE CAN REACH

About 25,500 firms

1 in 5 run their jobs over WhatsApp

SOM · OUR 3-YEAR GOAL

About 500 firms

2% of those: the ones we aim to win

Circles not to scale. 128,000 = 102,657 construction firms (DOSM Economic Census 2023) plus about 25,000 others (our assumption). The 1 in 5 and the 2% are assumptions too.

FUTURE PLAN

Prove it. Expand it. *Let it learn.*

Q1 2027

Prove

- First paying event company
- Official WhatsApp Business API
- Measure changes caught and hours saved

2027

Expand

- More event firms
- Renovation, catering, printing, wholesale
- Orbyte add-on

WITH ENOUGH DATA

Learn

- Train on every approved change
- Suggest extras before the client asks
- Today: counts from past jobs, e.g. 4 of 4 demo dinners ran overtime

Impact KPI: agreed changes that reach the bill, from an estimated 75% to 95% · final-bill time cut by half

SDG 8.2 · 8.3: small firms keep the revenue they have earned.

OUR ASK

Catch it. Confirm it. *Sync it.*

A *four-week pilot* with three to five event companies. Know a firm that runs its jobs over WhatsApp? Introduce us.

TEAM VBALL

VC Vincent Chew Wen Jie

SE Steven Elnathan Meliala

LW Looi Wan Wei

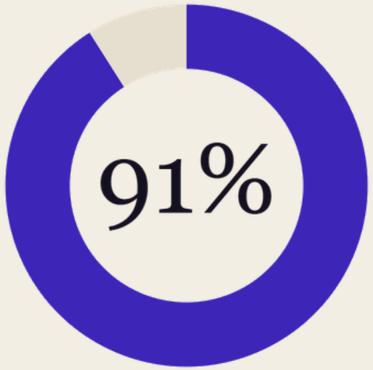
Orbyte Smart SME Challenge · Domain 2 · SDG 8.2 · 8.3

Scan for the code

github.com/Anonys-sky/SMErgers · Apache 2.0 · Live demo: sm-ergers-1.replit.app

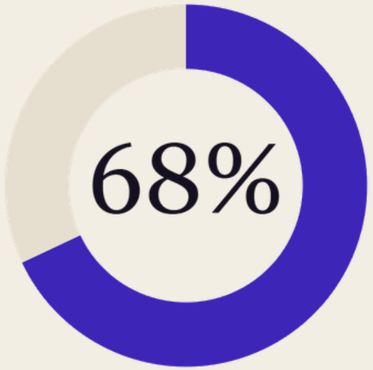
DOES THE READER WORK?

On chats it had never seen, the AI reader spotted *91% of the changes.*



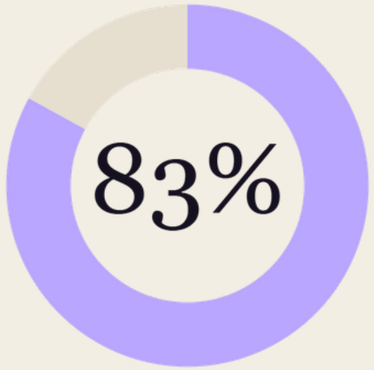
Spots the change

59 of 65 changes found



Fully right

44 of 65: change, price and status



After our fixes*

54 of 65, same chats

Plain rules alone: 75% spotted, 32% fully right. So the AI reads, code checks every price, and a person approves.

7 chats, 65 changes, written by outside agents who never saw our code; answer key written before scoring. *Light donut: same chats after we fixed its mistakes, so not independent. Every run: docs/results.md.

APPENDIX C

Hard questions, plain answers

“Isn’t it just an invoice maker?”

No. It finds and confirms what was agreed. Invoicing stays in accounting software.

“Why not just write it down?”

Deals happen mid-chat at 11 pm. We read the chat they already use.

“How accurate is it?”

On 65 changes in chats written by outsiders, the AI reader spotted 91% and got 68% fully right; plain rules 75% and 32%. After our fixes, 83% fully right on those same chats (not independent). Small synthetic samples, so a pilot comes next.

“Is the AI reader proven?”

Not on real chats yet. The pilot tests that.

“Is reading WhatsApp allowed?”

Through Meta’s official API, in groups the business creates, with consent.

“What if the client never confirms?”

It stays off the bill, on the director’s list. Nothing is billed silently.

“Will owners pay RM100?”

The pilot tests it. Break-even is 3% of extras recovered.

“What if a message is misread?”

Unclear items go to a person. Nothing reaches the bill without a check and the client’s yes.

Rehearse these with a teammate playing a hostile judge.

RESULT IN OUR TEST

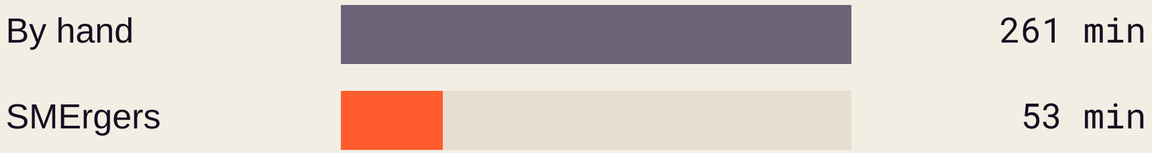
Every agreed change *reached the bill.*



48 of 48 changes

across 11 chats. Starting point: about 75% (our estimate).

TIME TO BUILD THE FINAL BILL



80% less time

Modelled from stated assumptions. A timed test with people is next.

The reader proposes, a person finishes: 13 fixed, 2 added and 3 rejected along the way.

A person checked every change. Reproduce with python src/benefit.py; assumptions are at the top of that file. Report: docs/benefit.md.

APPENDIX D

Limits, stated plainly

NOT YET PROVEN

Real chats. Everything is synthetic.

Real owners paying. A 7-person survey, no interviews yet.

The time saved. Modelled, timed test pending.

Official WhatsApp groups. Needs a verified business.

KNOWN WEAKNESSES

The AI reader can call an unconfirmed change ready.

It can price a "price later" item.

Data is in memory and resets on restart.

Sign-in is simulated; roles are enforced.

SAFEGUARD

A person checks every change. The client confirms. The log records both.

Sources for every number: docs/results.md and docs/benefit.md in the repository.

SOLUTION

People stay in control

Who	Can	Cannot
Project manager	Check, add a missing price, reject	Approve big items or discounts
Project director	Approve RM500 or more, decide discounts	Confirm for the client
Client	Confirm, or dispute with a reason	See the team’s pages

HUMAN APPROVAL

Nothing is billed without the PM’s check and the client’s yes.

CHECKED ON THE SERVER

Not just hidden buttons. A client link can’t approve its own change.

ACTIVITY LOG

Who, role, what, when. Each entry is chained, so an edit shows.

Sign-in is simulated, as the brief allows. The RM500 limit is a setting.

APPENDIX B

Boundaries, privacy and tools

INSIDE THE BRIEF

- No payments or filings
- No legal or financial advice
- Bill is a draft, not an invoice
- Synthetic data only
- Sign-in simulated; roles still enforced

PDPA, IN REAL USE

- Tell people why the group is read
- Keep only messages tied to changes
- Limit access by role
- Set retention; honour requests
- Legal review before any pilot

TOOLS DECLARED

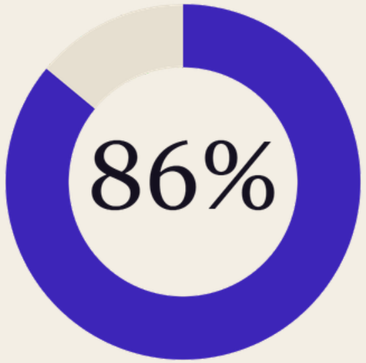
- Replit: build and host
- GitHub: public, Apache 2.0
- Claude, Replit Agent: AI coding help
- Llama 3.1 8B: the AI reader
- WhatsApp Cloud API: design
- RunPod: GPU that runs the AI reader (no training)

WhatsApp, honestly: Meta’s official API is built for business-to-customer chats and needs a verified business account. An unofficial client (whatsapp-web.js) was used only for team testing, never in the product.

Boundaries from the Domain 2 brief, section 6. Not legal advice.

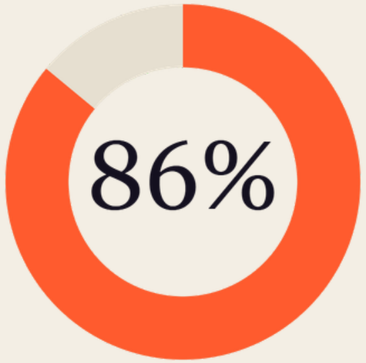
WHAT 7 EVENT PROFESSIONALS SAID

86% find it useful *and pick a paid plan.*



Find it useful

6 of 7 rated it 4 or 5 out of 5. Average 4.3.



Pick a paid plan

6 of 7 chose RM100 or RM250 a month as fair.

THEIR PAIN, IN SHORT

> Clients add things, then deny agreeing.
> OWNER, 100+ JOBS A YEAR

> They do not know extra time costs money.
> HOTEL EVENT OPERATIONS

> Last-minute adds land on finance.
> FINANCE, EVENT COMPANY

One owner has already asked to meet us. That is our first pilot conversation.

7 responses, 5 event companies, 31 Oct 2026, our network, self-reported. The question described our product. Words shortened from the free-text answers. Table: docs/survey.md.

APPENDIX A

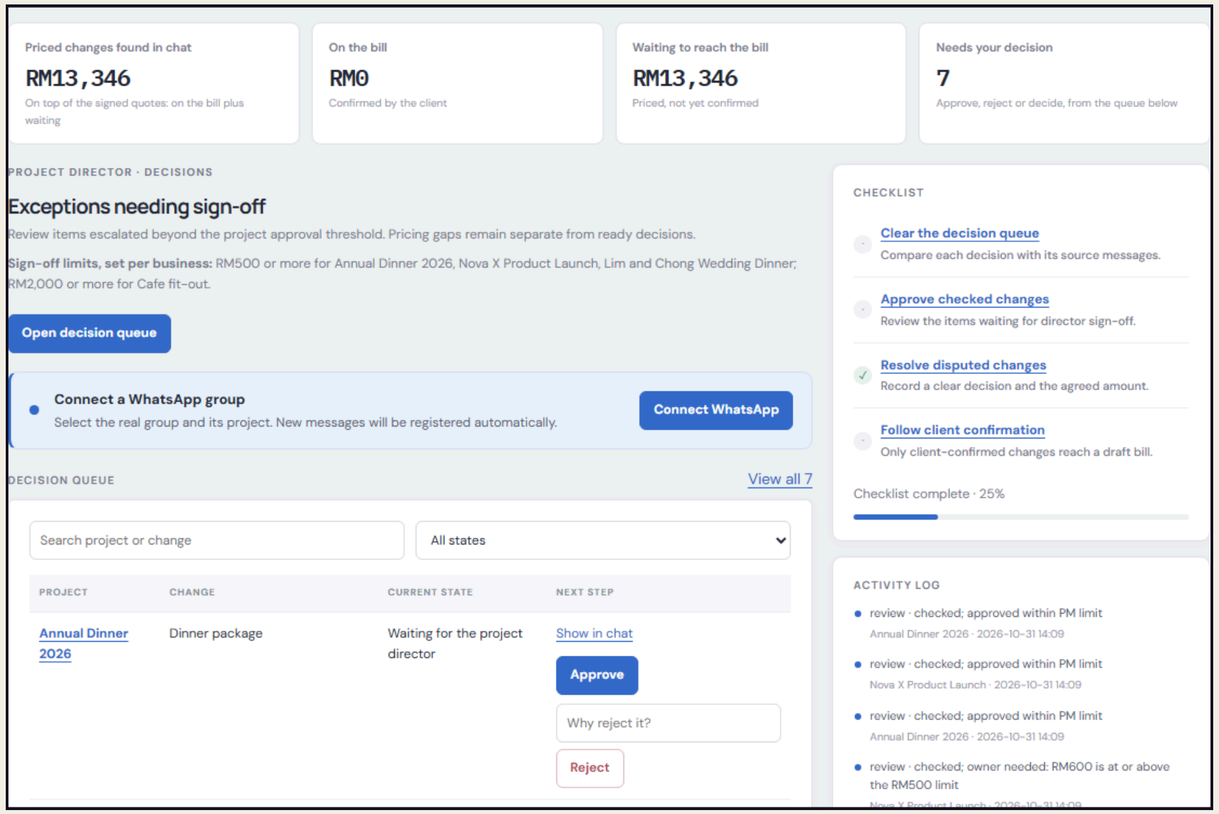
Every number: source, confidence, caveat

Number	Source	Confidence	Caveat
102,657 construction and renovation firms (market input)	DOSM Economic Census 2023	High	Whole sector, not only renovation
About 25,000 other project firms	Our assumption	Low	No public event-company count
20% run jobs over WhatsApp	Our assumption	Low	To measure in the pilot
35 messages, 10 changes, RM3,860 (11.2%) on one job	Synthetic dinner chat (data/chats)	Synthetic	One job; real leakage unknown
20 admin hours at RM25	Our assumption	Low	Owners to confirm
Competitor features (grid)	Vendor pages, search results	Medium	Front pages only; not tested
Changes at “almost every event”	Team member’s experience	Medium	One person, one company
Blind chats: AI spotted 59 of 65 and got 44 fully right; rules 21	docs/results.md, Runs 2, 5, 7 and 8	Low	Synthetic chats by AI agents; 65 changes. 54 fully right after fixes, same chats (not independent)

A widely repeated “52% of projects see scope creep” figure could not be traced to its source, so it is not used. Checked 21 to 24 Oct 2026.
dosm.gov.my · rentman.io · helloendless.com · workflowmax.com

THE OWNER'S VIEW

Built for the person who *signs off.*



1 **The money at a glance**
Found in chat, on the bill, waiting to reach it.

2 **Decide in one row**
Evidence one click away. Approve, or reject with a reason.

3 **Your limit, your rule**
Each business sets the amount from which the director signs off.

Screenshot of the running app, synthetic data.

LIVE DEMO · AS THE OWNER

Monday morning: *what needs me?*

1

Messages arrive

A real WhatsApp group on a test number is connected. New messages are read within seconds and logged.

2

I see the money

Found in chat, on the bill, waiting. What needs my decision, with its evidence.

3

Approve or reject

One row, one click. A rejection needs a reason and tells the PM.

4

Client confirms

From a link, no login. Quote v2 goes back to the group.

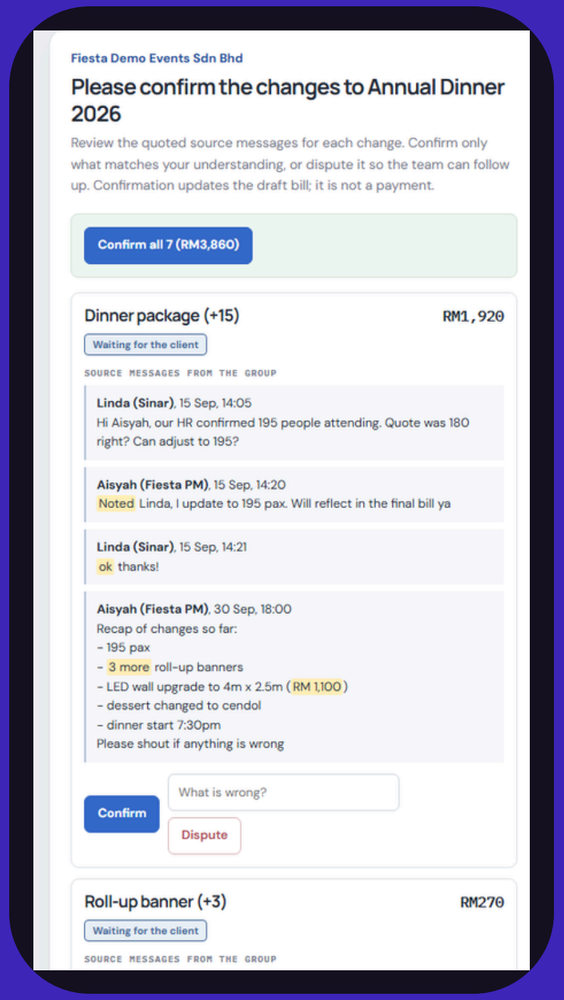
5

Prove it

Activity log: who did what, when. An edit would show.

Then *a second business*, with its own sign-off limit.

Test numbers, sample business data. Live site: sm-ergers-1.replit.app · Backup video if the network fails.



1 · Client taps the link

SOLUTION

The client confirms. Everyone stays in sync.

Annual Dinner · WhatsApp group

SMErgers

Confirmed by Sinar Demo Trading Sdn Bhd:

- Dinner package (+15): RM1,920
- LED wall 3m x 2m, changed to 4m x 2.5m: RM1,100
- Emcee overtime (+1): RM300
- and 4 more

Quote Q-2026-041 is now version 2, total RM38,390.

Version 1

RM34,530

Version 2

RM38,390

2 · The group is told. Demo: synthetic group. Production: WhatsApp Business API.

Screenshot and message are from the working prototype, synthetic data.

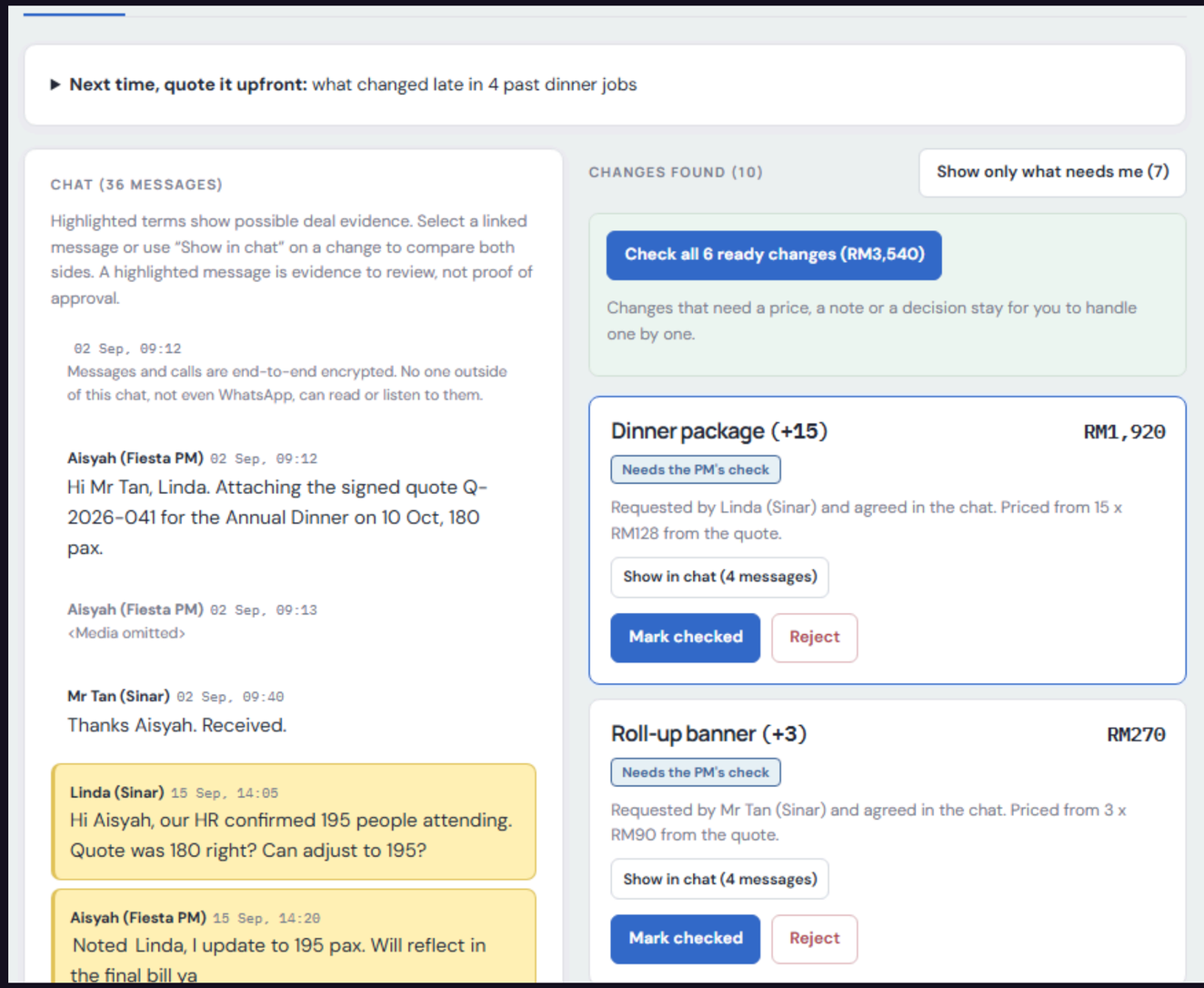
SOLUTION

The project manager's view

1 **Know what waits for you**
RM3,540 agreed, 7 changes to check

2 **Every change has proof**
The client's own words light up

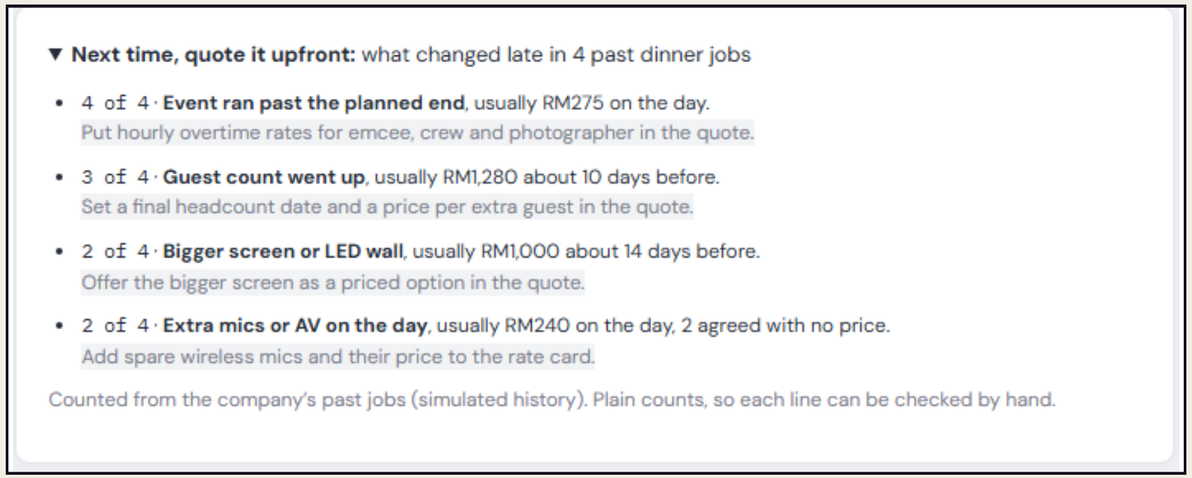
3 **Check six in one click**
Unpriced items wait for a person



Screenshot of the working prototype, synthetic data.

TECHNICAL APPROACH

It learns from every job



▼ **Next time, quote it upfront:** what changed late in 4 past dinner jobs

- 4 of 4 · **Event ran past the planned end**, usually RM275 on the day.
  Put hourly overtime rates for emcee, crew and photographer in the quote.
- 3 of 4 · **Guest count went up**, usually RM1,280 about 10 days before.
  Set a final headcount date and a price per extra guest in the quote.
- 2 of 4 · **Bigger screen or LED wall**, usually RM1,000 about 14 days before.
  Offer the bigger screen as a priced option in the quote.
- 2 of 4 · **Extra mics or AV on the day**, usually RM240 on the day, 2 agreed with no price.
  Add spare wireless mics and their price to the rate card.

Counted from the company's past jobs (simulated history). Plain counts, so each line can be checked by hand.

SYNTHETIC DEMO HISTORY

4 of 4

past dinners in our demo ran past the planned end.
Put overtime in the quote.

NOW

Plain counts from past jobs. Fewer last-minute decisions.

NEXT

Every approval is a labelled example. After a pilot, it trains a model.

Built on 10 synthetic past jobs of one demo company. A model is trained on real approvals after a pilot.

SINCE THE PRELIMINARY ROUND

What we built *and why.*

A demo you step through by hand	→	**A real WhatsApp group (test number) that triggers by itself**
Reader only; a miss was a dead end	→	**A person can add what it missed; the director signs it off**
Reject was a click	→	**Reject needs a reason, is logged, and tells the other side**
One approval limit for everyone	→	**Each business sets its own**
Benefit estimated	→	**Benefit counted from the engine and the log**
One blind test, rules looked fine	→	**A second one: the rules fall to 22%, the AI reader holds at 75%**

Next: a timed bill test with real people, a pilot with 3 to 5 event companies on the official WhatsApp API, and Orbyte approvals and eSign.

352 automated tests pass. Every run is logged in docs/results.md.

WHAT YOU CAN CHECK

Ask us to prove it. *It is all live.*

Does it trigger by itself?	Connect a WhatsApp group: new messages are read within seconds and every arrival is logged.
Can I approve or reject?	Yes. Rejecting needs a reason, stays off the bill and tells the other side.
Who can see and act?	Switch role: PM, director, client, finance. The server refuses the wrong role.
Is there a real record?	Activity log, each entry chained to the one before. An edit shows.
Is there a second business?	Café renovation: its own quote file and its own approval limit. Add a third in one form.
Does it work on chats you wrote?	Send a message in the group yourself, or import a chat. Results are in the repository.

Everything on this slide can be opened in the live app.

BUSINESS

Market size



TAM · PROJECT-BASED SMES

≈128,000 firms × RM1,200 a year

Renovation and contracting 102,657 firms (DOSM 2023) + about 25,000 events, agencies, print (assumption)

SAM · RUN JOBS OVER WHATSAPP

≈25,500 firms (20% of TAM, assumption)

3 or more staff, jobs billed by project

SOM · WITHIN THREE YEARS

≈510 firms (2% of SAM, assumption)

Event companies first, through Orbyte and associations

Sources and every assumption: appendix A. No event-company count exists in public data.

BUSINESS

Does it pay for itself?

1 · EXTRAS A YEAR

RM38,600

RM3,860 on our dinner, times 10 dinners a year

→

2 · WON BACK

RM9,650

We win back one quarter of those extras

−

3 · SUBSCRIPTION

RM1,200

Confirm tier: RM100 a month, for 12 months

=

4 · OWNER GAINS

RM8,450

7.0 times what they paid

WIN BACK MORE OR LESS

Won back	RM a year	Return (ROI)	With admin time (ROV)
10%	3,860	2.2×	3.6×
25%	9,650	7.0×	8.5×
50%	19,300	15.1×	16.5×

Return = (money won back − RM1,200) ÷ RM1,200. With admin time also counts 20 hours saved at RM25.

3% is all it takes. Winning back 3% of extras pays for the whole subscription.

One simulated job, so the pilot measures the real rate. Hours and hourly rate are assumptions: appendix A.

WHO PAYS AND HOW BIG

The owner pays, because *it pays back.*

PER COMPANY, PER MONTH · CLIENTS NEVER PAY

Capture	Free	One job at a time. Find and list changes.
Confirm	RM100	Approvals, client links, quote updates.
Settle	RM250	All jobs, draft bills, CSV, suggestions.

MARKET, BOTTOM-UP · EVERY ASSUMPTION FLAGGED

TAM RM153m

About 128,000 project-based small firms

SAM RM31m

20% scope jobs over WhatsApp (assumption)

SOM RM0.6m

2% of that in three years (assumption)

The test: win back about 3% of the extras and the subscription has paid for itself.

Illustrative; the pilot measures the real rate. Survey (7 people): 4 chose the RM250 tier, 2 chose RM100. DOSM Census 2023 for the firm count; the rest are assumptions.

BUSINESS

Business model

B2B · freemium subscription (SaaS)

CAPTURE

RM0

always free

One job at a time. Find and list every change.

CONFIRM

RM100

per company, per month

Approvals, client links, quote updates to the group.

SETTLE

RM250

per company, per month

All jobs. Draft bills, CSV, suggestions, log export.

Free to start. It pays for itself with the first extra it recovers.

B2B: businesses pay. Their clients never do.

Channel: Orbyte, then event associations. Option: usage credits.

Launch prices, validated with owners in the pilot.

BUSINESS

Who it is for

Micro and small businesses that run on WhatsApp



START HERE

Event companies

Dinners, launches, weddings. Changes fly in WhatsApp.

NEXT

Renovation, contractors

Same flow: extras agreed on site, then argued.

THEN

Agencies, print, IT

Any team that bills by the job and scopes in chat.

BUYER

The owner. Loses money to unbilled extras.

DAILY USER

The project manager. Chases details across chats.

TOUCHPOINT

The client. Taps a link. No login.

Photo: Unsplash.

11 PM · A TUESDAY · THE EVENT TEAM'S GROUP CHAT

Client

"Ok, go ahead."

Three words. A bigger LED wall. *Nobody wrote it down.*

Synthetic example, from a team member's experience at an event company.